

CHAPTER 2: STRUCTURE OF ATOM (JEE FOCUS)

Topics Covered:

- Subatomic particles
- Atomic models
- Bohr model
- Dual nature of matter
- Quantum mechanical model
- Quantum numbers
- Electronic configuration

Core Concepts:

- Electron (−), Proton (+), Neutron (0)
- Z = protons = electrons (neutral atom)
- A = protons + neutrons
- Isotopes → same Z , different A
- Rutherford → nucleus + empty space
- Bohr → quantized orbits (H-like only)
- Energy $\propto -1/n^2$
- Dual nature → wave + particle
- Orbital → probability region
- Quantum numbers define electron state
- n, l, m_l, m_s → full description
- Pauli → max 2 electrons/orbital
- Hund → max unpaired first
- Aufbau → lowest energy fills first

Key Formulas:

- $E_n = -13.6 Z^2 / n^2$ (eV)
- $\Delta E = E_f - E_i$
- $\lambda = h / mv$
- $E = h\nu$
- $c = \nu\lambda$
- $1/\lambda = RZ^2 (1/n_1^2 - 1/n_2^2)$
- Max electrons = $2n^2$

Important Patterns:

- H-like → apply Bohr directly
- Spectral lines → depend on ΔE
- Higher n → closer energy levels
- Electronic order → $n + l$ rule
- 4s fills before 3d, empties first
- Cr, Cu → stability exceptions
- Quantum numbers → range check
- Orbitals in shell = n^2

Common Mistakes:

- Using Bohr for multi-electron atoms
- Wrong orbital filling order

- Ignoring Cr, Cu exceptions
- Invalid quantum number sets
- Confusing orbit vs orbital
- Sign mistakes in energy
- Forgetting spin $\pm 1/2$

Examples (Important Cases):

- Energy of He^+ , Li^{2+}
- Lyman/Balmer wavelength
- Valid quantum number sets
- Electronic configuration of Cr, Cu
- Orbitals in $n = 3, 4$

Exam Tips:

- Identify H-like or not first
- Memorize orbital order
- Use elimination in MCQs
- Visualize spectral transitions
- Practice exceptions repeatedly
- Keep constants ready
- Treat as rule-based chapter